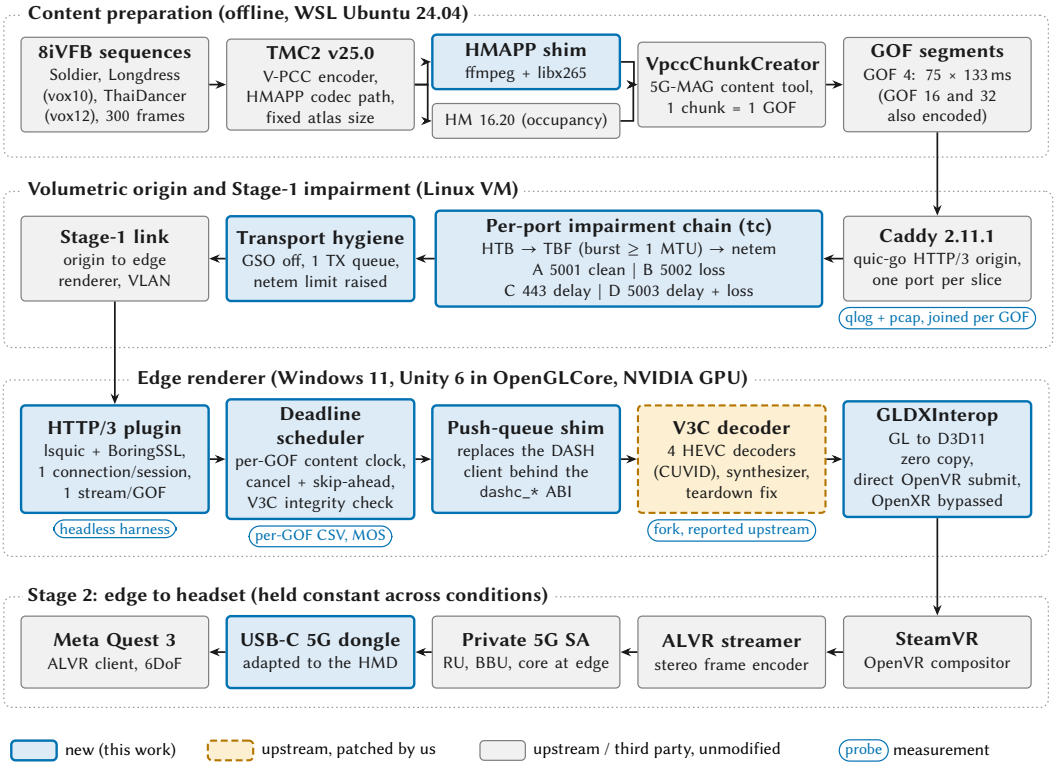
Content preparation (offline, WSL Ubuntu 24.04)
8iVFB sequences
Soldier, Longdress (vox10), ThaiDancer (vox12), 300 frames
TMC2 v25.0
V-PCC encoder, HMAPP codec path, fixed atlas size
HMAPP shim
ffmpeg + libx265
HM 16.20 (occupancy)
VpccChunkCreator
5G-MAG content tool, 1 chunk = 1 GOF
GOF segments
GOF 4: 75 × 133 ms (GOF 16 and 32 also encoded)
Volumetric origin and Stage-1 impairment (Linux VM)
Caddy 2.11.1
quic-go HTTP/3 origin, one port per slice
qlog + pcap, joined per GOF
Per-port impairment chain (tc)
HTB → TBF (burst ≥ 1 MTU) → netem
A 5001 clean | B 5002 loss
C 443 delay | D 5003 delay + loss
Transport hygiene
GSO off, 1 TX queue, netem limit raised
Stage-1 link
origin to edge renderer, VLAN
Edge renderer (Windows 11, Unity 6 in OpenGLCore, NVIDIA GPU)
HTTP/3 plugin
lsquic + BoringSSL, 1 connection/session, 1 stream/GOF
headless harness
Deadline scheduler
per-GOF content clock, cancel + skip-ahead, V3C integrity check
per-GOF CSV, MOS
Push-queue shim
replaces the DASH client behind the dashc_* ABI
V3C decoder
4 HEVC decoders (CUVID), synthesizer, teardown fix
fork, reported upstream
GLDXInterop
GL to D3D11 zero copy, direct OpenVR submit, OpenXR bypassed
Stage 2: edge to headset (held constant across conditions)
SteamVR
OpenVR compositor
ALVR streamer
stereo frame encoder
Private 5G SA
RU, BBU, core at edge
USB-C 5G dongle
adapted to the HMD
Meta Quest 3
ALVR client, 6DoF
new (this work)
upstream, patched by us
upstream / third party, unmodified
probe
measurement